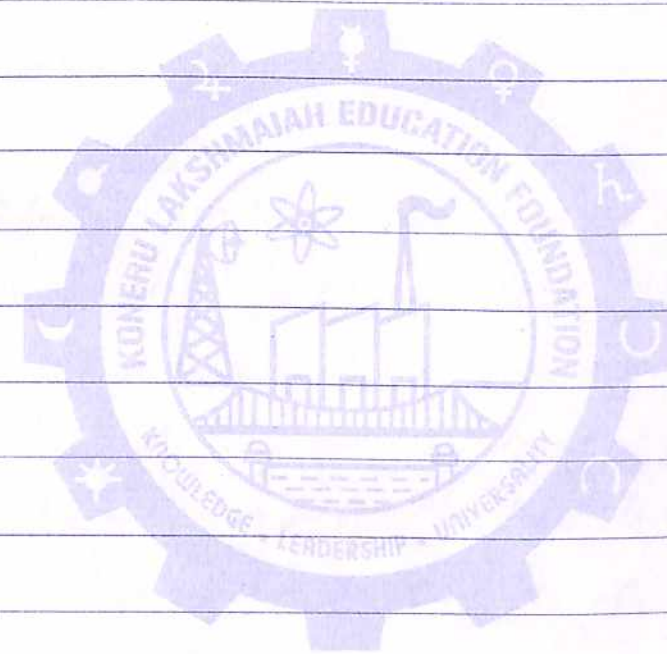


### **INSTRUCTIONS TO THE CANDIDATES**

- ❖ Do not use sketch pens/ high –lighter pens/ color pencils etc., in underlining / highlighting certain points in the answer or while drawing sketches etc.,
- ❖ Check the Answer booklet thoroughly. The defective answer booklet may be returned to the invigilator before starting the examination.
- ❖ Ensure that the OMR Barcode sheet is properly secured to the answer booklet.
- ❖ Do not write on or tamper the Barcode. If you do so, your booklet will not be evaluated.
- ❖ The candidate is prohibited from writing his/ her NAME / ADDRESS/ PHONE NUMBER / EMAIL ID / Religion or any other matter on any part of the answer booklet / drawing sheet/ graph sheet/ addressing the examiner in any manner whatsoever in the answer booklet. The candidate is not supposed to tear a page in full or part of the booklet.  
**Non-adherence with this instruction shall result in candidate being booked under malpractice.**
- ❖ Answer should be written on both sides of the paper. Candidate should write in all 25 lines in each page. It is not necessary to begin each answer in a fresh page.
- ❖ Candidate should write only question number in the margin.
- ❖ Candidate should return the answer booklet to the invigilator before leaving the examination hall.
- ❖ **ADDITIONAL SUPPLEMENT OF ANSWER SHEETS WILL NOT BE ISSUED.**

### **INSTRUCTIONS TO THE EXAMINERS**

- ❖ Invigilators should ensure that the above instructions are strictly followed by the candidates.
- ❖ After verifying the particulars on the OMR sheet, invigilator should sign in the box specified.
- ❖ Do not crumple or cut the sides of the sheets.
- ❖ Do not write, mark or damage the timing tracks or Barcode.



6 A

A company is concerned about the environmental impact of its Mining operations. Can blockchain technology help track and manage the energy consumption and waste generated during Mining processes?

In simple words, yes it can be monitored. For starters, when a miner is connected to the block chain it itself becomes a node. And as a miner, it needs to solve complex algorithms for it to be rewarded. As the miner needs to be provided with electricity, if we compare both the electricity consumed and the crypto value mined we can see if it's self sustainable or not.

6 B

The double spending problem in Blockchain and list out the methods to avoid this using Blockchain technology.

Double spending is a process in Block chain technology this couldn't happen in a centralised system.

It is when a single user says that he will reward the miner for his work done, but this is said to multiple people. So the first person who completes it will be rewarded and the other are left with nothing.

4 A

The Message Authentication Code (MAC) can satisfy Authentication and Confidentiality only.

The MAC can be used to encrypt and decrypt the plain text. It is useful and can be utilized by various things.

4 B

A Cryptographic Hash Function.

$$P = 5 \text{ and } Q = 3$$

$$n = 15$$

$$\phi n = (5-1)(3-1)$$

$$= 8$$

$$\gcd(8, 3) = 1$$

$$PT = 7$$

$$7^3 \cdot 15$$

2

D

Double Spending

Double Spending is not possible in a centralized system.

IT is when a person assigns a work to multiple people showing a reward same to all.

The miner who completes the task first shall be rewarded by and all the rest is not rewarded.

A

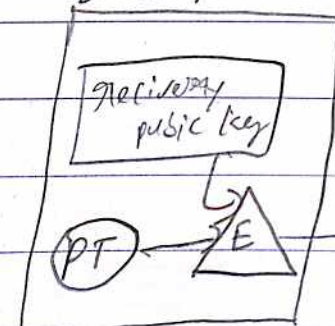
Asymmetric Cryptosystem. (Public Key)

In Asymmetric there are 2 keys involved in the process, one is used for encryption and the other is used for decryption.

Encryption  $\rightarrow$  Public Key

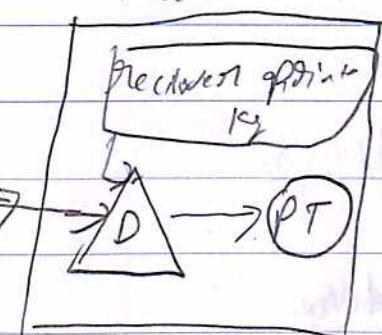
Decryption  $\rightarrow$  Private Key

Sender.



CT

Receiver.



B

SHA-256.

The plain text and cipher text will be 256 characters long.

$$2^8 = 256$$

$$4 \times 4 \times 4 \times 4 \\ 2+2+2+2+2+2+2+2 \\ 16 \quad 16$$

D

Blockchain

- Hidden Identity
- Decentralized
- Open to everyone
- Transparent.

1 C

## Attacks.

### Active

• Passive attacks

↳ eavesdropping

• Active attacks

↳ Man in middle.

↳ Data modification

↳ DoS Attack

↳ privilege escalation.

1 E

## Smart contract

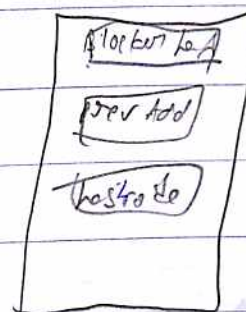
A contract is bond/agreement between 2 users.

A

A smart contract is a contract that makes sure the deal is completed for the ~~payment~~ transition to happen.

1 F

## Block.



### Block header.

↳ Merkle tree

↳ Address of previous

↳ value.

Hash Code → hash code of its own.









